the collection or publication of data. If the measurement errors are systematic, in general, auxiliary equations can be specified to capture these errors. This unit will focus only on the impact of random measurement errors on the regression model.

12.2 CONSEQUENCES OF ERRORS IN VARIABLES

Most of the published data or summary information contain errors of summarizing or misrepresenting by informer. When these data are used, one of the assumptions of classical lease-squares method is violated. In this case, the classical least-squares estimators will be biased even when sample size increases, which is alternatively known as asymptotic biases or inconsistency.

Under the classical assumptions the ordinary least squares (OLS) estimators are best linear unbiased. One of the major underlying assumptions is the interdependence of repressors from the disturbance term. If this condition does not bold, OLS estimators are biased and inconsistent. This statement may be illustrated by simple errors in variables.

We discuss about the consequences in the following, if the error appears in the measurement of dependent variable, independent variable or both.

12.2.1 Measurement Error in Y

Let us assume that only the dependent variable contains error of measurement. Assume that the true regression model (written in deviation form) is

$$y_i = \beta x_i + \varepsilon_r \quad \ldots (12.1)$$

where ε_i represents errors associated with the specification of the model (the effects of omitted variables, etc.).

Assume that the variable y^*, instead of y, is obtained in the measurement process such that

$$y_i^* = y_i + u_i, \quad \ldots (12.2)$$

The measurement error u_i is not associated with the regressor. Thus we have

$$\mathrm{Cov}(u_i, x_i) = 0$$

The regression model is estimated with y^* as the dependent variable, with non account being taken of the fact that y^* is not an accurate measure of y. therefore, instead 0 estimating Eq. (12.1), we estimate

$$y^* = y_i + u_i = (\beta x_i + \varepsilon_i) + u_i$$

or, $$y^* = \beta x_i + (\varepsilon_i + u_i)$$

or, $$y^* = \beta x_i + e_i \quad \ldots (12.3)$$

where e_i is a composite error term, containing the population disturbance term ε_i (which may be called the error term in the equation) and the measurement error term u_i.

For simplicity let us assume that $E(\varepsilon_i) = E(u_i) = 0$, $Cov(x_i, \varepsilon_i) = 0$ (which is the assumption of classical linear regression), and $Cov(x_i, u_i) = 0$, i.e., the errors of measurement in y^* are uncorrelated with x_i, and $Cov(\varepsilon_i, u_i) = 0$, the equation error and measurement error are uncorrelated.

With these assumptions, it can be shown that β estimated from either (12.1) or (12.3) will be an unbiased estimator of the true β. Thus, the errors of measurement in dependent variable y^*, do not destroy the unbiased property of the OLS estimators.

However, the variances and standard errors of β estimated from (12.1) and (12.3) will be different because, employing the usual formula, we obtain

For (12.1), $$Var(\hat{\beta}) = \frac{\sigma_\varepsilon^2}{\sum x_i^2} \qquad \ldots (12.4)$$

For (12.3), $$Var(\hat{\beta}) = \frac{\sigma_\varepsilon^2}{\sum x_i^2} = \frac{\sigma_\varepsilon^2 + \sigma_u^2}{\sum x_i^2} = \frac{\sigma_\varepsilon^2}{\sum x_i^2} + \frac{\sigma_u^2}{\sum x_i^2} \qquad \ldots (12.5)$$

Obviously, the variance given at (12.5) is larger than the variance given at (12.4). Therefore, although the errors of measurement in the dependent variable still give unbiased estimates of the parameters and their variables, the estimated variances are now larger than the case where there are no such errors of measurement.

12.2.2 Measurement Error in X

We have taken the true regression model deviation form to be

$$y_i = \beta x_i + \varepsilon_i \qquad \ldots (12.1)$$

Now let us assume that explanatory variable x_i is measured with error and the observed value becomes x_i^* such that

$$x_i^* = x_i + \nu_i$$

or, $$x_i = x_i^* - \nu_i$$

Substituting the value of x_i in (12.1), we have

$$\begin{aligned} y_i &= \beta(x_i^* - \nu_i) + \varepsilon_i \\ &= \beta x_i^* - \beta\nu_i + \varepsilon_i \\ &= \beta x_i^* + (\varepsilon_i - \beta\nu_i) \\ &= \beta x_i^* + (w_i) \end{aligned}$$

where $w_i = \varepsilon_i - \beta\nu_i$, We have assumed earlier that the measurement error in x is normally distributed with zero mean, it has no serial correlation, and it is uncorrelated with ε_i. However, we can no longer assume that the composite error term w_i is independent of the explanatory variable x_i^*.

$$Cov(w_i, x_i^*) = E[w_i - E(w_i)]\,[x_i^* - E(x_i^*)]$$

$$= E[w_i - 0]\,[(x_i^* - x_i)]$$
$$= E[\varepsilon_i - \beta\nu_i]\,\nu_i$$
$$= E[(\varepsilon_i - \nu_i) - \beta\nu_i^2]$$
$$= 0 - \beta\sigma_v^2$$
$$= -\beta\sigma_v^2 \qquad \ldots (12.6)$$

The composite error term w_i has mean zero as $E(w_i) = E[\varepsilon_i - \beta\nu_i] = E(\varepsilon_i) - \beta E(\nu_i) = 0$.

Thus, the explanatory variable and the error term are correlated, which violates the crucial assumption of the classical linear regression model that the explanatory variable is uncorrelated with the stochastic disturbance term. If this assumption is violated, then OLS estimates are not only biased but also inconsistent; they remain biased asymptotically. Let us now show this.

Instead of being able to estimate true relationship (12.1), we are forced to replace x_i by x_i^*. Now the OLS estimator is

$$\hat{\beta} = \frac{\sum x_i^* y_i}{\sum x_i^{*2}}$$
$$= \frac{\sum (x_i + \nu_i) y_i}{\sum (x_i + \nu_i)^2}$$
$$= \frac{\sum (x_i + \nu_i)(\beta x_i + \varepsilon_i)}{\sum (x_i + \nu_i)^2}$$
$$= \frac{\beta \sum x_i^2 + \beta \sum x_i \nu_i + \sum x_i \varepsilon_i + \sum \nu_i \varepsilon_i}{\sum x_i^2 + \sum \nu_i^2 + 2\sum x_i \nu_i}$$

From above we find that $\hat{\beta}$ is a biased estimate as $E(\hat{\beta}) \neq \beta$. Let us see the asymptotic properties of $\hat{\beta}$.

Since ν_i and ε_i are stochastic and are uncorrelated with each other as well as with x_i, we can say that

$$p\lim \hat{\beta} = p\lim \frac{\beta \sum x_i^2}{\sum x_i^2 + \sum \nu_i^2}$$
$$= \frac{\beta \mathrm{Var}(x)}{\mathrm{Var}(x) + \sigma_v^2} \text{ (Since } x_i = X_i - \bar{x} \text{ by definition)}$$
$$= \frac{\beta \sigma_{x2}^2}{\sigma_x^2 + \sigma_v^2}$$
$$= \beta \frac{\beta \sigma_x^2}{\sigma_x^2 + \sigma_v^2}$$

$$= \beta \frac{1}{1 + \frac{\sigma_v^2}{\sigma_x^2}} < \beta \qquad \text{… (12.7)}$$

Thus, $\hat{\beta}$ is biased even for an infinite sample and $\hat{\beta}$ is an inconsistent estimator of β.

12.2.3 Measurement Errors in both X and Y

Now let us assume that both X and Y have errors of measurement. The true model is as before

$$y_i = \beta x_i + \varepsilon_i$$

Since X and Y have errors of measurement, we observe x_i^* and y_i^* instead of x_i and y_i, such that

$$y_i^* = Y_i + u_i, \qquad x_i^* = x_i + \nu_i,$$

where u_i and ν_i present the errors in the values of y_i and x_i respectively. We make the following assumptions about the error terms:

(i) There is no correlation between the error term and corresponding variable, i.e.,

$$E(y_i, u_i) = 0, \; E(x_i, y_i) = 0$$

(i) There is no correlation between error of one variable and measurement of the other variable, i.e.,

$$E(u_i, x_i) = 0 \;\; E(\nu_i, y_i) = 0$$

(i) There is no correlation between errors in measurement of both the variables, i.e.,

$$E(u_i, \nu_i) = 0$$

On the basis of the above assumptions, our estimated regression equation will be

$$y_i = \beta x_i$$

or, $$y_i - u_i = \beta(x_i^* - \nu_i)$$

or, $$y_i^* = \beta x_i^* - \beta\nu_i + u_i$$

or, $$y_i^* = \beta x_i^* + (\mu_i - \beta\nu_i) \qquad \text{…(12.8)}$$

Equation (12.8) shows that if we model y^* as a function of x^*, the transformed disturbance contains the measurement error in β. We then write the OLS estimator $'\beta'$ as

$$\hat{\beta} = \frac{\sum x_i^* y_i^*}{\sum x_i^{*2}}$$

By substituting the values of x^* and y^* in terms of x and y we find that

$$\hat{\beta} = \frac{\sum (x_i + \nu_i)(y_i + u_i)}{\sum (x_i + \nu_i)^2}$$

$$= \frac{\sum (x_i + \nu_i)(\beta x_i + u_i)}{\sum (x_i + \nu_i)^2}$$

$$= \frac{\beta \sum x_i^2 + \beta \sum x_i \nu_i + \sum x_i u_i + \sum \nu_i u_i}{\sum x_i^2 + \sum \nu_i^2 + 2\sum x_i \nu_i}$$

From above we observe that $E(\hat{\beta}) \neq \beta$

$$p\lim \hat{\beta} = p\lim \frac{\beta \sum x_i^2}{\sum x_i^2 + \sum \nu_i^2}$$

$$= \frac{\beta \mathrm{Var}(x)}{\mathrm{Var}(x) + \sigma_v^2}$$

$$= \frac{\beta \sigma_x^2}{\sigma_x^2 + \sigma_v^2}$$

$$= \beta \frac{\sigma_x^2}{\sigma_x^2 + \sigma_v^2}$$

$$= \beta \frac{1}{1 + \frac{\sigma_v^2}{\sigma_x^2}} < \beta \qquad \ldots (12.9)$$

Thus, $\hat{\beta}$ will not be a consistent estimator of β. The presence of measurement error of the type in question will lead to an underestimate of the true regression parameter if ordinary least squares are used.

Example 12.1

A Specific illustration of this problem in economics is the permanent income theory of the consumption function, which is an errors-in-variables model. According to this model measured income Y^* has a permanent (true) component. Y and a transitory component, Y_ε such that

$$Y^* = Y + Y_\varepsilon \qquad \ldots (12.10)$$

Measured consumption expenditure, C^*, similarly, has a permanent (true) component, C, and a transitory component, C_ε such that

$$C^* = C + C_\varepsilon \qquad \ldots (12.11)$$

As you know form MEC-002, Block 4, the permanent consumption (C) is proportional to permanent income (Y). Due to transitory components, however, the average propensity to consume declines as income of a household increases. Let us analyse this issue through errors in variables.

The transitory components represent accidental or chance factors, including cyclical variations. They can be treated as the errors in measuring income and consumption. It is usually assumed that errors, on the average, vanish:

$$E(Y_\varepsilon) = E(C_\varepsilon) = 0 \qquad \ldots (12.12)$$

It is also usually assumed that the errors are correlated neither with each other nor with the permanent components:

$$Cov(Y, Y_\varepsilon) = Cov(C, C_\varepsilon) = Cov(Y_\varepsilon, C_\varepsilon) = 0 \quad \ldots (12.13)$$

According to permanent-income hypothesis,

$$C = \beta_1 Y + \beta_2 + u \quad \ldots (12.14)$$

that is, permanent consumption is a linear function of permanent income, where β_1 and β_2 are constant parameters and u is a stochastic disturbance term. Notice that the parameter β_1 in (12.14) represents marginal propensity to consume (MPC). Combining (12.14) with (12.10) and (12.11) yields

$$C^* = \beta_1 Y^* + \beta_2 + (\nu - \beta_1 Y_\varepsilon), \qquad \text{where } \nu = u + C_\varepsilon \quad \ldots (12.15)$$

Equation (12.15) is a simple linear regression model of the same form as (12.8). Then,

$$p\lim \hat{\beta}_1 < \beta_1 \qquad \text{(as we know from (12.9))}$$

Since the ratio β_1 is less than unity, we observe that the above term is less than the true parameter.

Thus, the estimated marginal propensity to consume, $\hat{\beta}_1$, even in the probability limit, systematically underestimates the true β_1. The greater the variance of transitory income, the greater will be the underestimating. Only when the variance of transitory (error) component of income is zero, the least squares estimator of β_1 is a consistent estimator. To solve this problem, it is necessary to construct a measure of permanent income Y before estimating the consumption function.

Check Your Progress 1

1) Explain the concept of errors in variable. What are its consequences?

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2) State whether the following statements are true or false?

a) Errors in variables lead to estimates of the regression coefficients that are biased.

b) The presence of measurement error in an equation leads to an underestimate of the true regression parameter if ordinary least squares are used.

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3) Explain briefly why measurement error in the explanatory variables leads to biased and inconsistent parameter estimates while measurement error in the dependent variable does not.

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12.3 INSTRUMENTAL VARIABLES METHOD

The problem of errors in the measurement of variables in regression models is quite important, yet econometricians do not have much to offer in the way of useful solutions. As a general rule, we tend to pass over the problem of measurement error, hoping that the errors are small enough which will not destroy the validity of the estimation procedure. One technique which is available and can solve the measurement error problem is the technique of instrumental variables estimation. We briefly outline the concept of instrumental variables because it is likely to be useful with measurement errors.

The method of instrumental variables involves the search for new variable Z which is highly correlated with the independent variable X and at the same time uncorrelated with the error term in the equation (as well as the errors of measurement of both variables). In practice, we are concerned with the consistency of parameter estimates and therefore concentrate on the relationship between the variable Z and the remaining variables in the model when the sample size gets large. We define the random variable Z to be an instrument if the following conditions are met.

1) The correlations between Z and $\varepsilon, u, \text{and } \nu$, respectively, in the regression approach zero, as the sample size gets large.
2) The correlation between Z and X is nonzero, as the sample size gets large.

We simply select one instrument (or combination of instruments) that has the highest correlation with the X variable.

Assuming for the moment that such a variable can be found, we can alter the least squares regression procedure to obtain estimated parameters that are consistent. Unfortunately, there is no guarantee that the estimation process will yield unbiased parameter estimates. To simplify the matter, let us consider the case of measurement errors in the independent variable such $y_i = \beta x_i + \varepsilon_i$ and only x is measured with error (as $x^* = x + \nu$). In order to solve the problem we take the regression equation $y_i = \beta z_i + \varepsilon_i$ Where Z is the instrumental variable. The instrumental variables estimator of the regression slope in the above model is

$$\hat{\beta} = \frac{\sum y_i z_i}{\sum x_i^* z_i} \qquad \dots (12.16)$$

The choice of the particular slope formula is made so that the resulting estimator will be consistent. To see this, we can derive the relationship between the instrumental-variables estimator and the true slope parameter as follows:

$$\hat{\beta} = \frac{\sum y_i z_i}{\sum x_i^* z_i}$$

$$= \frac{\sum \left(\beta x_i^* + \varepsilon_i^*\right) z_i}{\sum x_i^* z_i}$$

$$= \frac{\beta \sum x_i^* z_i + \sum z_i \varepsilon_i^*}{\sum x_i^* z_i}$$

$$= \beta + \frac{\sum z_i \varepsilon_i^*}{\sum x_i^* z_i}$$

Clearly, the choice of Z as an instrument guarantees that $\hat{\beta}$ will approach β as the sample size gets large $\left[\text{Cov}\left(z, \varepsilon^*\right) \text{approaches } 0\right]$ and will therefore be a consistent estimator of β. Remember that the variable x_i^* in (12.16) was not replaced by z_i in the denominator of instrumental variables estimator, as the estimator $\frac{\sum y_i z_i}{\sum z_i^2}$ does not yield a consistent estimator of β. Let us show this,

$$\hat{\beta} = \frac{\sum y_i z_i}{\sum z_i^2}$$

$$= \frac{\sum y_i z_i}{\sum z_i^2}$$

$$= \frac{\sum \left(\beta x_i^* + \varepsilon_i^*\right) z_i}{\sum z_i^2}$$

$$= \frac{\beta \sum x_i^* z_i + \sum z_i \varepsilon_i^*}{\sum z_i^2}$$

$$= \frac{\beta \sum \left(x_i + v_i\right) z_i + \sum z_i \varepsilon_i^*}{\sum z_i^2}$$

$$= \frac{\beta \sum x_i z_i + \beta \sum z_i v_i + \sum z_i \varepsilon_i^*}{\sum z_i^2}$$

$$p\lim \hat{\beta} = p\lim \frac{\beta \sum x_i z_i}{\sum z_i^2}$$

$$= \beta \frac{\text{Cov}\left(x_i z_i\right)}{\sigma_z^2} \neq \beta$$

The above shows that $\frac{\sum y_i z_i}{\sum z_i^2}$ will not yield a consistent estimator of β.

The technique of instrumental variables appears to provide a simple solution to a difficult problem. We have defined an estimation technique, which yields consistent estimates if we can find an appropriate instrument.

A few concluding comments may be instructive.

i) First, the OLS estimation technique is actually a special case of instrumental variables. This follows because in the classical regression model X is correlated with the error term and because X is perfectly correlated with itself.

ii) Second, if we generalize the measurement error problem to errors in more than one independent variable, one instrument is needed to replace each of the designated independent variables.

iii) Finally, we repeat that instrumental-variables estimation guarantees consistent estimation but does not guarantee unbiased estimation.

12.4 TEST OF MEASUREMENT ERRORS

Suppose we are estimating the two-variable regression model

$$y_i = \beta x_i + \varepsilon_i \qquad \text{… (12.1)}$$

There is a possibility that x might be measured with error.

If $x_i = x_i^* - \nu_i$, then the actual least squares regression would be

$$y_i = \beta\left(x_i^* - \nu_i\right) + \varepsilon_i$$

$$= \beta x_i^* - \beta\nu_i + \varepsilon_i$$

$$= \beta x_i^* + \left(\varepsilon_i - \beta\nu_i\right)$$

$$= \beta x_i^* + \varepsilon_i^* \qquad \text{… (12.17)}$$

where $\varepsilon_i^* = \varepsilon_i - \beta\nu_i$

If x is measured with error, we have seen that consistent estimator of β can be obtained by using an instrument z which is correlated with x^* but uncorrelated with ε and ν. Suppose the relationship between z and x^* is given by

$$x_i^* = \gamma z_i + w_i \qquad \text{… (12.18)}$$

When (12.18) is estimated using least squares, we obtain

$$x_i^* = \hat{\gamma} z_i$$

or $$x_i^* = \hat{x}_i^* + \hat{w}_i \qquad \text{… (12.19)}$$

where $\hat{w}_i$ are the regression residuals. Substituting the value of eq. (12.19) into eq. (12.17) we have

$$y_i = \beta\left(\hat{x}_i + \hat{w}_i\right)$$

$$y_i = \beta\hat{x}_i^* + \beta\hat{w}_i + \varepsilon_i^* \qquad \text{… (12.20)}$$

Whether or not there is measurement error, the coefficient of $\hat{x}_i^*$ will be consistently estimated by ordinary least squares, since

$$p\lim \frac{\sum \hat{x}_i^* \varepsilon_i^*}{N} = p\lim \frac{\hat{\gamma}\sum z_i\left(\varepsilon_i - \beta\nu_i\right)}{N} = 0.$$

In fact, the least-squares estimator of the coefficient of $\hat{x}_i^*$ in eq. (12.20) is identically equal to the instrumental-variables estimator, which is given by

$$\hat{\beta} = \frac{\sum y_i z_i}{\sum x_i^* z_i}$$

To look at the coefficient of the variable $\hat{w}_i$, note that

$$\text{p}\lim \frac{\sum \hat{w}_i \varepsilon_i^*}{N} = \text{p}\lim \frac{\sum \left(x_i^* - \hat{\gamma} z_i\right)\left(\varepsilon_i - \beta \nu_i\right)}{N}$$

$$= \text{p}\lim \frac{-\beta \sum \nu_i \left(x_i + \nu_i\right)}{N}$$

$$= -\beta \sigma_v^2$$

When there is no measurement error, $\sigma_v^2 = 0$, so that OLS applied to eq. (12.20) will generate a consistent estimator of the coefficient of $\hat{w}_i$. However, when there is measurement error, the coefficient will be estimated inconsistently.

This suggests a relatively easy measurement error test. Let δ represent the coefficient of the variable $\hat{w}_i$ in eq. (12.20).

Substituting $\hat{x}_i^* = x_i^* - \hat{w}_i$, we get

$$y_i = \beta\left(x_i^* - \hat{w}_i\right) + \delta \hat{w}_i + \varepsilon_i^*$$

or $$y_i = \beta x_i^* + (\delta - \beta)\hat{w}_i + \varepsilon_i^* \qquad \dots (12.21)$$

With no measurement error, $\delta = \beta$, so that the coefficient of $\hat{w}_i$ should equal zero. However, with measurement error, $\delta \neq \beta$, and the coefficient will (in general) be different from zero. We can test for measurement error by doing a simple two-stage procedure. First, we regress x^* on z to obtain the residuals $\hat{w}$. Then, we regress y on x^* and $\hat{w}$ and perform a t test on the coefficient of the $\hat{w}$ variable. If we are concerned with measurement error in more than one variable of a multiple regression model, an equivalent F test could be applied.

The test just described is a special case of a more general test for specification error proposed by Hausman. The Hausman specification test relies on the fact that under the null hypothesis the ordinary least-squares estimator of the parameters of the original eq. (12.17) is consistent and (for large samples) efficient, but is inconsistent if the alternative hypothesis is true. However, the instrumental-variables estimator [test least-squares estimator of eq. (12.20)] is consistent whether or not the null hypothesis is true, although it is inefficient if the null hypothesis is not valid.

The Hausman specification test is as follows: If there are two estimators $\hat{\beta}_1$ and $\hat{\beta}_2$ that converge to the true value β under the null but converge to different values under the alternative, the null hypothesis can be verified by testing whether the probability limit of the difference of the two estimators is zero.

Example 12.2

The expenditure of states and local governments (EXP) in US vary substantially by state and region. Among the important variables that explain difference in spending levels are federal grants-in-aid (AID), the income of states (INC), and